

x is equivalent to all int >= 0.
c=2n.

Chain length n=1, c=2

$$2^c - 1 + 2^{c+1}x, (2^c - 1 + 2^{c+1}x)/2, 3(2^{c-1}) - 1 + 3x2^c, 9^0 2^c + 9(2^{c-1})x, 9^0 2^c + 9(2^{c-2})x$$

Chain length n=2, c=4

$$2^c - 1 + 2^{c+1}x, ((2^c - 1) + 2^{c+1}x)/2, 3(2^{c-1}) - 1 + 3x2^c, ((3(2^{c-1}) - 1) + 3x2^c)/2, 2^{c-2} - 1 + 2^{c+1} + 9x2^{c-1}, (2^{c-2} - 1 + 2^{c+1} + 9x2^{c-1})/2, 3(2^{c-3}) - 1 + 3(2^c) + 27x2^{c-2}, 9^0 2^{c-2} + 9^1 2^{c-2} + 9^2 2^{c-2} + 9^0 2^{c-3}x, 9^0 2^{c-2} + 9^1 2^{c-2} + 9^2 2^{c-2} + 9^2 2^{c-4}x$$

Chain length n=3, c=6

$$2^c - 1 + 2^{c+1}x, ((2^c - 1) + 2^{c+1}x)/2, 3(2^{c-1}) - 1 + 3x2^c, ((3(2^{c-1}) - 1) + 3x2^c)/2, 2^{c-2} - 1 + 2^{c+1} + 9x2^{c-1}, (2^{c-2} - 1 + 2^{c+1} + 9x2^{c-1})/2, 3(2^{c-3}) - 1 + 3(2^c) + 27x2^{c-2}, (3(2^{c-3}) - 1 + 3(2^c) + 27x2^{c-2})/2, 2^{c-4} - 1 + 2^{c-1} + 9(2^{c-1}) + 81x2^{c-3}, 3(2^{c-5}) - 1 + 3(2^{c-2}) + 27(2^{c-2}) + 243x(2^{c-4}), 9^0 2^2 + 9^1 2^2 + 9^2 2^2 + 9^3 2^2 + 9^2 2^2 + 9^1 2^2 + 9^1 2^2 + 9^0 2^2 + 9^2 2^2 + 9^3 2^2 + 9^2 2^2 + 9^2 2^2 + 9^3 x$$